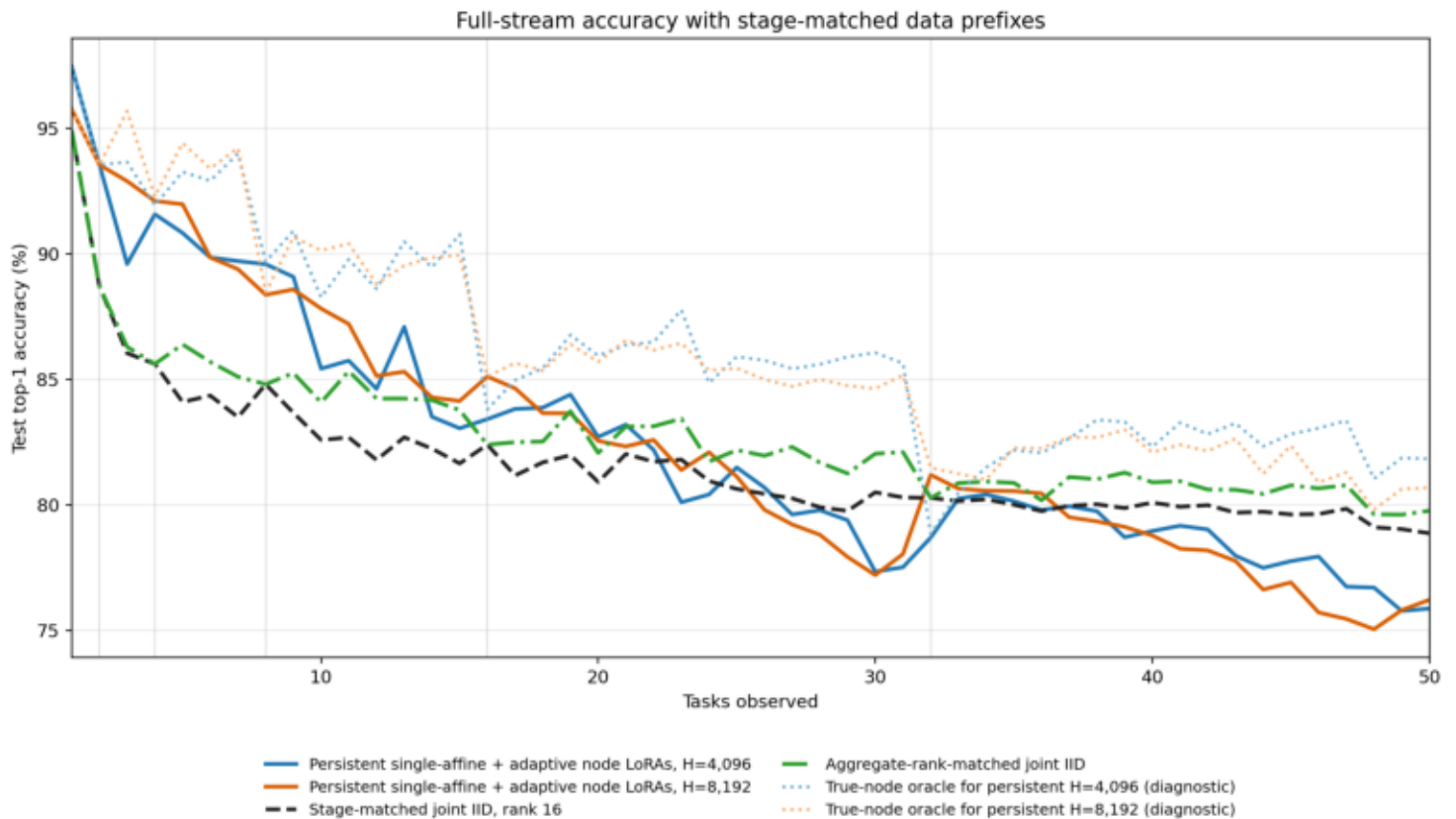


ImageNet-R-50 persistent single-affine frontier

Full 50-task extension of the H=4,096 and H=8,192 adaptive-LoRA conditions

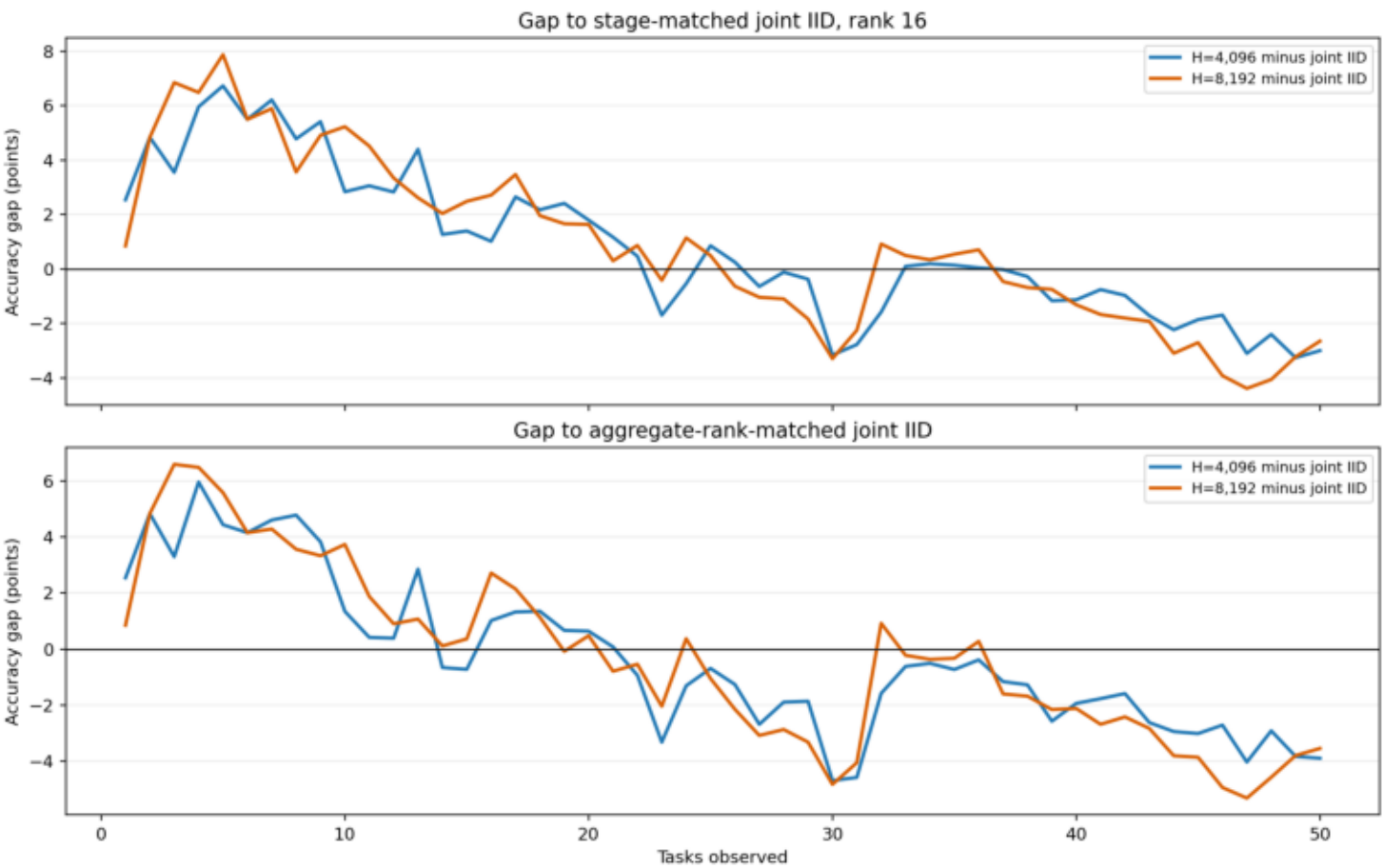
At task 50, H=4,096 reaches 75.867% and H=8,192 reaches 76.217%. The corresponding rank-16 and aggregate-rank joint-IID references are 78.867% and 79.767%. Across all 50 stages, incremental accuracy is 82.434% for H=4,096 and 82.452% for H=8,192, versus 81.630% and 82.759% for the two joint-IID curves.



The adaptive curves use only one affine layer over node-specific pre-classifier vectors. Their dotted true-node curves are label-aware diagnostics, not deployable conditions. The black control is fresh rank-16 joint IID at each prefix. The green control uses rank 16 times the number of live nodes.

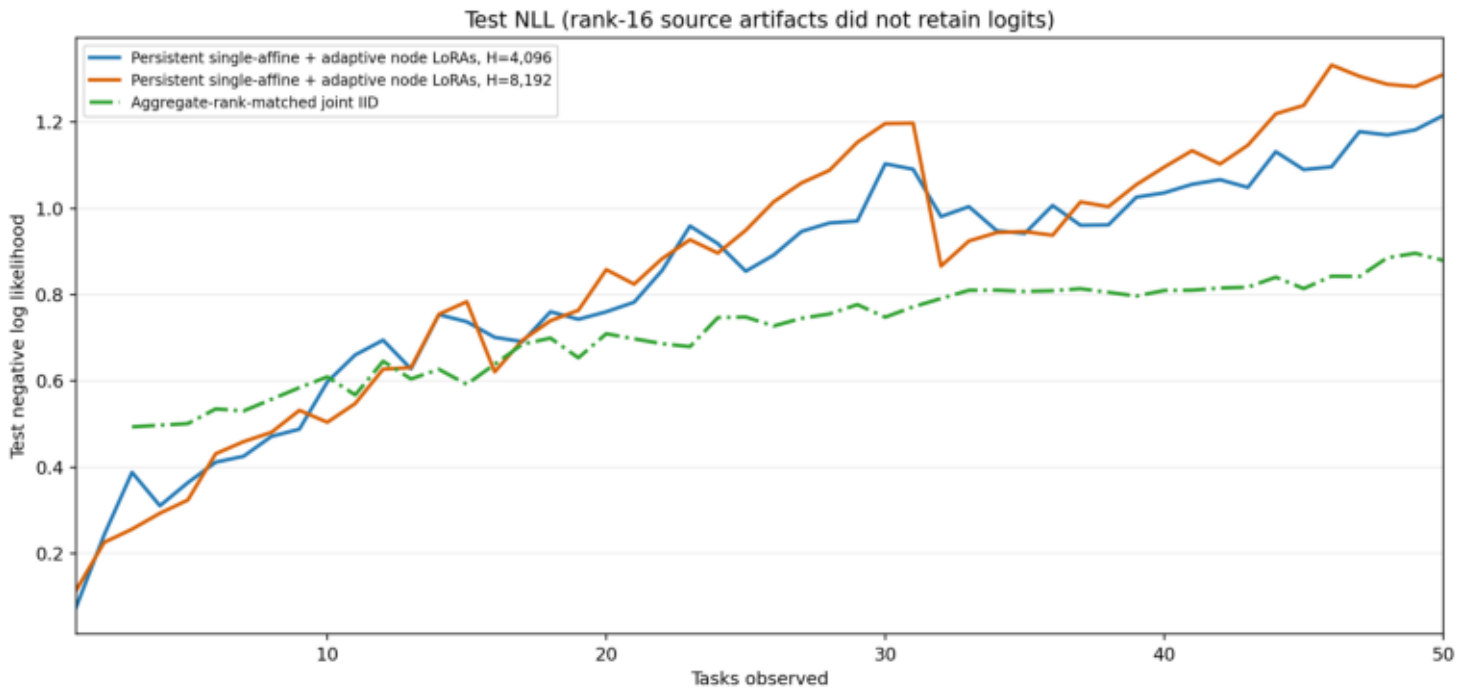
Selected stages and control gaps

Stage	Nodes	H=4,096	H=8,192	H4 oracle	H8 oracle	Joint r16	Rank-matched	Rank
1	1	97.46%	95.76%	97.46%	95.76%	94.92%	94.92%	16
2	1	93.55%	93.55%	93.55%	93.55%	88.71%	88.71%	16
4	1	91.58%	92.11%	91.93%	92.28%	85.61%	85.61%	16
8	1	89.59%	88.37%	89.68%	88.46%	84.80%	84.80%	16
16	1	83.41%	85.11%	83.80%	85.11%	82.40%	82.40%	16
31	5	77.52%	78.04%	85.64%	85.17%	80.29%	82.10%	80
32	1	78.69%	81.19%	78.85%	81.47%	80.28%	80.28%	16
50	3	75.87%	76.22%	81.83%	80.68%	78.87%	79.77%	48



Positive gap means the persistent frontier is ahead of the named joint-IID reference. Power-of-two stages have one node; fragmented stages have two or more.

Likelihood and protocol



Rank-16 source artifacts retained hard predictions rather than logits, so rank-matched NLL is absent at stages 1, 2, 4, 8, 16, and 32. Accuracy remains available at all 50 stages.

Exact online update rule

Every arrival uses all current-task images plus a deterministic stage-keyed class-stratified historical draw. $H=4,096$ receives four passes and $H=8,192$ receives five. AdamW state and affine parameters carry forward. A node adapter and its moments carry only if its hierarchy-node hash survives; new leaves and consolidation parents start from authenticated source adapters. The base ViT and local classifiers never train.

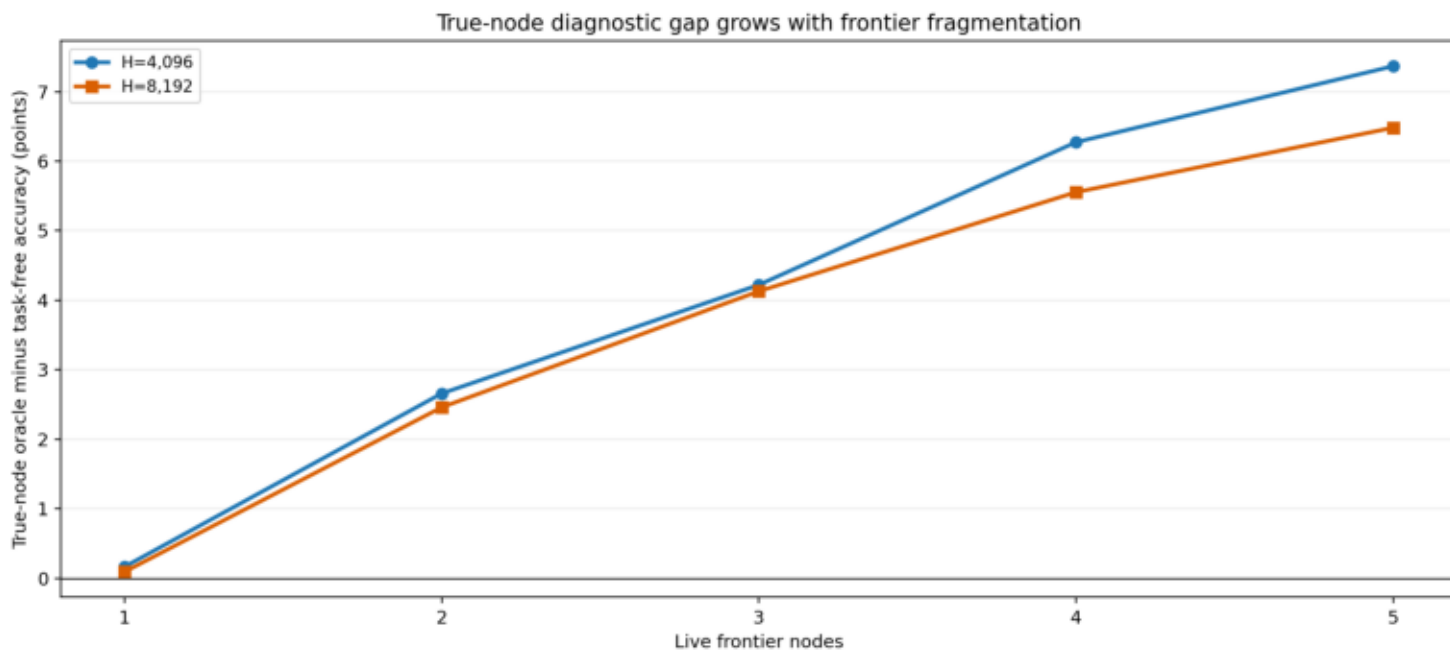
Comparison boundary

Stage-matched joint IID trains one fresh rank-16 adapter for five epochs on every prefix. Aggregate-rank-matched joint IID uses rank and alpha equal to 16 times popcount(stage). It matches total live LoRA rank, but not pretrained node state, multiple ViT paths, affine-head parameters, or deployment compute. Neither curve gates execution.

Provenance

Protocol 5f60e2f5c25e3318ba9c9b66d41ce8c159ba57a0f1a4fc524567c098b222d686
Result 4fa601c9b8e3cc87f07897f06ba51f88b75b7ecde1bf47ef01881526df4b4b58
Source hierarchy unchanged: True

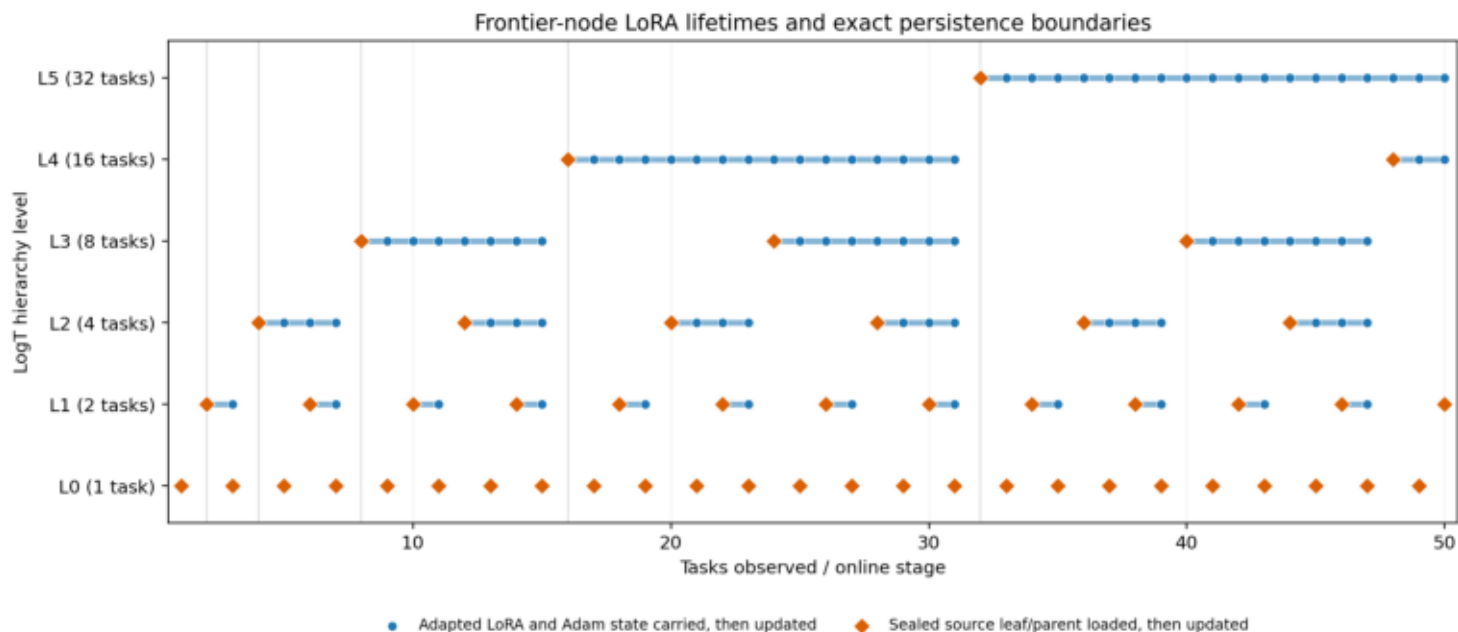
Fragmentation diagnosis



Nodes	Stages	H4 acc.	H4 gap	H8 acc.	H8 gap
1	6	89.05%	0.16	89.35%	0.09
2	15	83.76%	2.66	84.31%	2.46
3	18	81.39%	4.22	81.05%	4.13
4	9	79.10%	6.27	78.82%	5.55
5	2	77.13%	7.37	76.75%	6.48

The label-aware true-node oracle is diagnostic only. Its growing advantage as the number of independently adapted frontier nodes rises is consistent with cross-node competition. Because it also substitutes the frozen node-local classifier, it does not isolate routing alone.

Frontier-node adaptation lifetimes



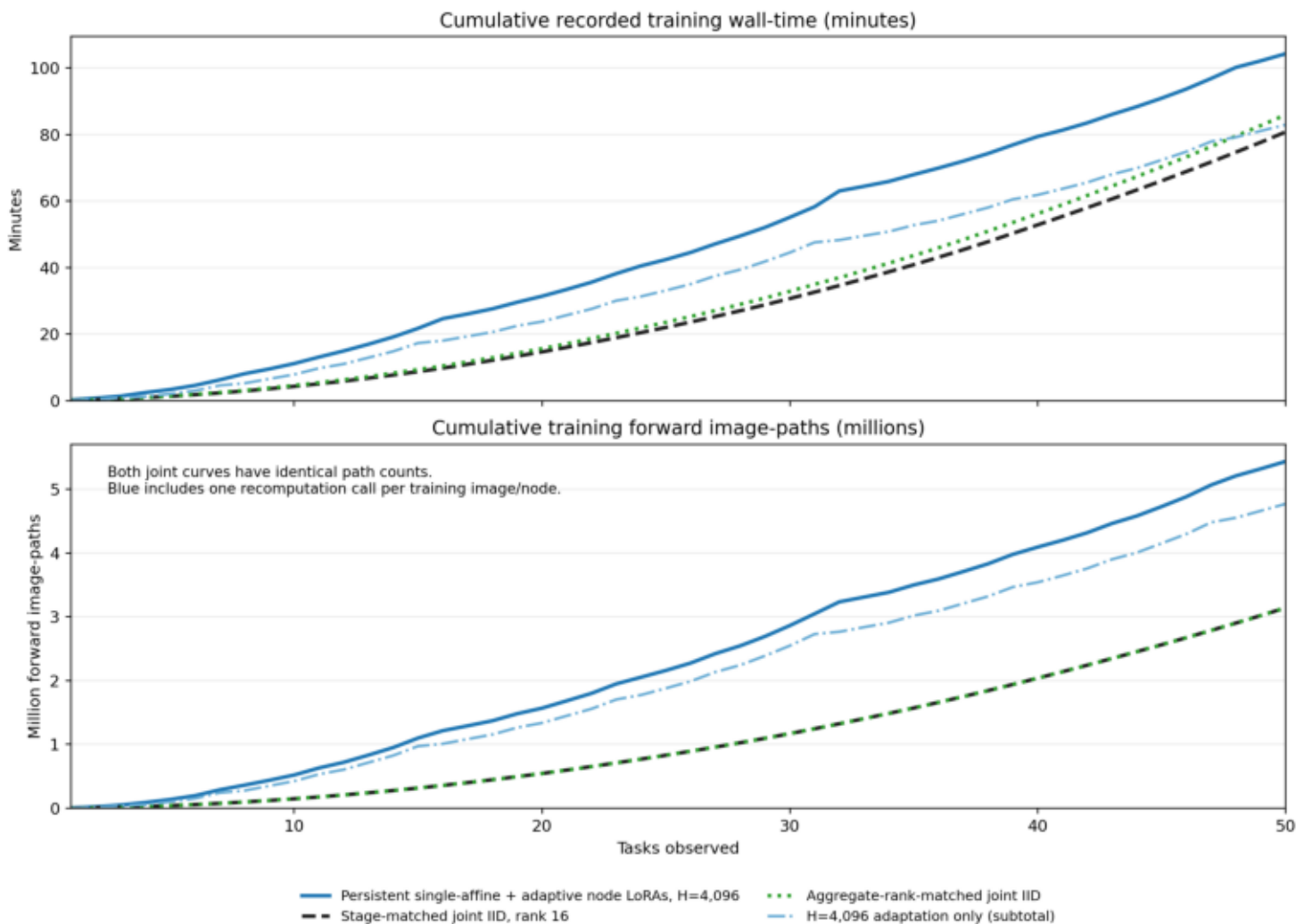
What persists

A blue lifetime continues only while the exact source hierarchy-node hash remains live. The next stage begins from that node's final adapted rank-16 LoRA, its 768-column affine block, and its named AdamW moments. All live LoRAs then update jointly on every current-plus-replay example.

What resets

An orange diamond is a new leaf or consolidation parent loaded from the sealed source hierarchy. A parent was independently retrained on the full union of its represented tasks; it does not inherit the online-adapted LoRAs or optimizer moments of retired children. Unaffected nodes continue normally, and old global affine-bias rows remain persistent.

Cumulative training cost



Training cost at 50 tasks

H=4,096 costs 1.29 times the rank-16 joint curve's recorded training time and 1.73 times its forward image-path count including checkpoint recomputation. The hierarchy subtotal includes all 50 leaves and 47 parents, including intermediate parents created and retired in the same arrival. The smaller asymptotic bound has not produced a wall-time saving at this horizon.

What is measured

One image-path is one image processed by one ViT with one installed adapter. Forward and backward counts are reconstructed from completed image-presentation counters and actual live-node counts. Recomputation is one additional forward invocation per image/node under activation checkpointing; it can stop early. These are path counts, not profiled FLOPs or equal-cost forward/backward operations. The affine head, optimizer, and differing LoRA ranks also affect wall time.

Resource accounting and scaling

Condition / component	Train min	Forward M	Recompute M	Backward M
H=4,096 total	104.23	3.051	2.385	3.051
H4 hierarchy subtotal	21.33	0.665	0.000	0.665
H4 adaptation subtotal	82.90	2.385	2.385	2.385
Joint IID, rank 16	80.72	3.140	0.000	3.140
Joint IID, rank matched	85.88	3.140	0.000	3.140

M = million image-paths. Recompute = activation checkpointing. H4 rows include two subtotals.

Wall-time and reuse

Curves sum recorded training-job wall times, including data loading and checkpoint writes inside those timers. They exclude setup, artifact validation, inter-job overhead, and evaluation. Source training is charged when its subtree becomes available. The imported final rank-16 joint model is charged its original training cost; each reused reference is counted once within each condition. These are alternative algorithm totals, not a sum of actual work in the v15 invocation.

Why ViT training is $O(T \log T)$

Let $P_t = 4(m_t + \min(4096, M_{t-1}))$, where m_t is the new task size and M_t is the seen training prefix. With $k_t = \text{popcount}(t)$, adaptive forwards are $A_T = \sum(k_t P_t)$. Each hierarchy example belongs to at most one five-epoch job per level, giving $S_T = 5 \sum_v(n_v)$. Total forward work including checkpoint recomputation is $S_T + 2A_T$; backward work is $S_T + A_T$. For fixed replay capacity, epochs, model size, and bounded task size, both are $O(T \log T)$. The joint curves each use $5 \sum_t(M_t)$ forward/backward pairs, which is $O(T^2)$ in model paths.

The entire benchmarking workflow has a larger bound

Repeated full-prefix tests add 471,565 forward image-paths and 6.06 measured minutes for H=4,096. Each joint curve adds 157,149 test paths. Rank-16 test time is 2.55 minutes; rank-matched test wall-time was not retained. Testing every growing prefix is $O(T^2 \log T)$ for the frontier. The replay sampler also scans all earlier identities at each arrival, giving at least quadratic CPU bookkeeping. The $O(T \log T)$ claim therefore applies to ViT training, not the whole runner. The affine output width is fixed at 200 here; unbounded class growth would require separate head-cost accounting.

Joint curves versus one final offline fit

Both joint curves refit at every prefix, so their base-model path counts coincide. The larger rank changes arithmetic per path, not the number of paths. A single final rank-16 offline fit cost 3.11 minutes and 120,000 forward/backward pairs; it provides only the final model and has linear training work in the total image count.